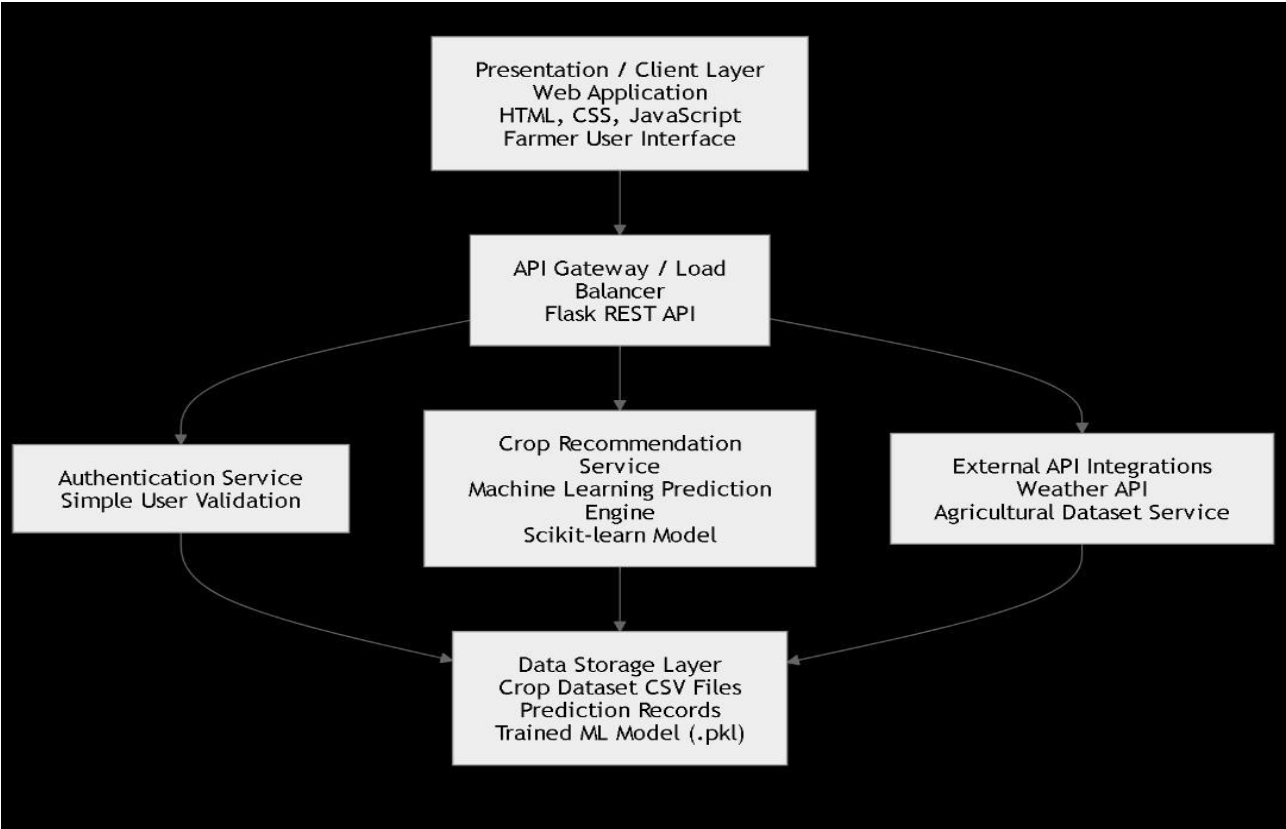


Solution Architecture

Date	29 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The solution architecture of the OptiCrop: Smart Agricultural Production Optimization Engine consists of three layers: Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides a web-based interface for farmers to enter soil parameters and view crop recommendations. The Application Layer contains the Flask API, machine learning prediction engine, authentication service, and optional external APIs. The Data Layer stores the agricultural dataset, trained machine learning model, and prediction records. This architecture ensures efficient, scalable, and accurate crop recommendation services for farmers.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface where farmers enter soil parameters and view crop recommendations.	HTML, CSS, JavaScript
API Gateway	Receives user requests and sends them to the machine learning service.	Flask REST API
Microservice	Processes soil data and predicts the best crop using machine learning algorithms.	Python, Scikit-learn, Pandas, NumPy
Database	Stores datasets, trained models, and prediction records.	CSV Files, Pickle (.pkl)